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JAVA PROJECT REPORT ON

Event Registration and Participant Management System

Submitted by

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Under the Guidance of

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DECLARATION

I hereby declare that the project entitled "**Event Registration and Participant Management System**" submitted in partial fulfillment of the requirements for the course **Java Programming** is a **record of my original work** carried out under the guidance of **MUGILAN R**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

_____

PREMKUMAR S

CERTIFICATE

This is to certify that the project entitled **"Event Registration and Participant Management System"** is a **bonafide work** carried out by **PREMKUMAR S B.Sc. IT „B" (2025-2028)**, **INFORMATION TECHNOLOGY**, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **Java Programming**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: Coimbatore

Date:

Submitted for the Viva-Voce Examination held on _____

Guide:
(Technical Trainer, KGCAS)

_____

Head of the Department
(Technical Head, KGCAS)

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1. INTRODUCTION

1.1 Objective

The main objective of the Event Registration and Participant Management System is to develop a simple and efficient application using Core Java that automates the process of event registration and participant management.

The specific objectives of this project are:

- To design and develop a Java-based event management system.
- To manage participant registration details efficiently.
- To store and track event participation records.
- To reduce manual errors in registration and data handling.
- To provide a user-friendly system for managing participants.
- To demonstrate the implementation of Core Java concepts such as classes, objects, methods, loops, and conditional statements.
- To generate a well-organized record of event participants.

1.2 Overview

The Event Registration and Participant Management System is a menu-driven application developed using the Java programming language. The system provides a digital platform where users can register participants for events and manage their details efficiently.

The system allows the user to enter participant information, assign them to events, and maintain a record of all participants. It helps organizers easily track who has registered and manage event data effectively.

This application eliminates manual record keeping and ensures faster and more accurate data management.

The system can be developed as a console-based application or a graphical user interface using Java Swing.

This project demonstrates the practical use of Java programming in real-world event management scenarios.

1.3 Features

The system includes the following features:

• Participant Registration – Allows entry of participant details such as name, ID, and contact information.

• Event Management – Stores details of different events.

• Participant Assignment – Assigns participants to specific events.

• Record Management – Maintains a list of all registered participants.

• Data Retrieval – Allows viewing of participant and event details.

• User-Friendly Interface – Simple menu-driven system for easy usage.

• Efficient Management – Reduces time and effort in managing event registrations.

2. PROGRAM ANALYSIS

Program analysis helps in understanding the problem and identifying system requirements.

The analysis phase focuses on identifying issues in manual systems and providing an efficient computerized solution.

2.1 Problem Definition

In many institutions and organizations, event registrations are handled manually using paper forms or spreadsheets.

Common problems include:

• Errors in recording participant details

• Difficulty in managing large numbers of participants

• Time-consuming registration process

• Lack of proper record management

• Difficulty in tracking participants for multiple events

Manual systems increase workload and may lead to confusion or data loss.

Therefore, there is a need for an automated system that can efficiently handle event registrations and participant records.

The Event Registration and Participant Management System is designed to solve these problems.

2.2 Project Overview

This system is a Java-based application that helps manage event registrations and participant information.

The system allows users to:

• Register participants

• Store participant details

• Assign participants to events

• View participant and event records

It uses Core Java concepts and can be implemented as a console or GUI application.

2.3 Scope and Limitations

Scope

• Managing participant registration

• Storing event details

• Assigning participants to events

• Viewing participant records

• Demonstrating Java programming concepts

Used in:

• Colleges

• Workshops

• Conferences

• Seminars

• Cultural events

Limitations

• No online registration feature

• Limited to small-scale event management

• No database integration (in basic version)

3. SYSTEM DESIGN

System design defines the structure and working of the system.

The system is designed as a menu-driven application with different modules.

3.1 Modules

1. Participant Registration Module

This module is used to enter and store participant details.

Example:

ID

Name

Phone Number

2. Event Management Module

This module stores event details such as:

Event Name

Event Date

Event Location

3. Participant Assignment Module

This module assigns participants to events.

Functions include:

• Selecting participant

• Selecting event

• Storing assignment

4. Record Viewing Module

This module displays:

• Participant details

• Event details

• Participant-event assignments

5. Exit Module

This module allows the user to exit the program safely.

---

4. METHODOLOGY

Methodology explains the steps followed in developing the project.

Stages include:

• Problem Analysis

• System Design

• Program Development

• Testing and Validation

• Result and Documentation

4.1 Problem Analysis

Manual registration systems are inefficient and error-prone.

Problems identified:

• Manual data entry errors

• Difficult record maintenance

• Slow registration process

• Poor data organization

4.2 System Design

The system is divided into modules:

• Participant Registration

• Event Management

• Assignment Module

• Record Viewing

• Exit Module

4.3 Program Development

The system is developed using Java.

Steps followed:

• Creating classes and objects

• Implementing menu logic

• Handling user input

• Storing participant and event data

• Displaying results

4.4 Testing and Validation

Test cases:

Test Case | Expected Result

Register Participant | Details stored correctly

Add Event | Event added successfully

Assign Participant | Assignment works correctly

View Records | Data displayed correctly

4.5 Result and Documentation

The system successfully stores participant and event data and displays records accurately.

Documentation includes input/output screenshots.

4.6 Algorithm

Start the program

Display menu

Register participant

Add event

Assign participant to event

Store details

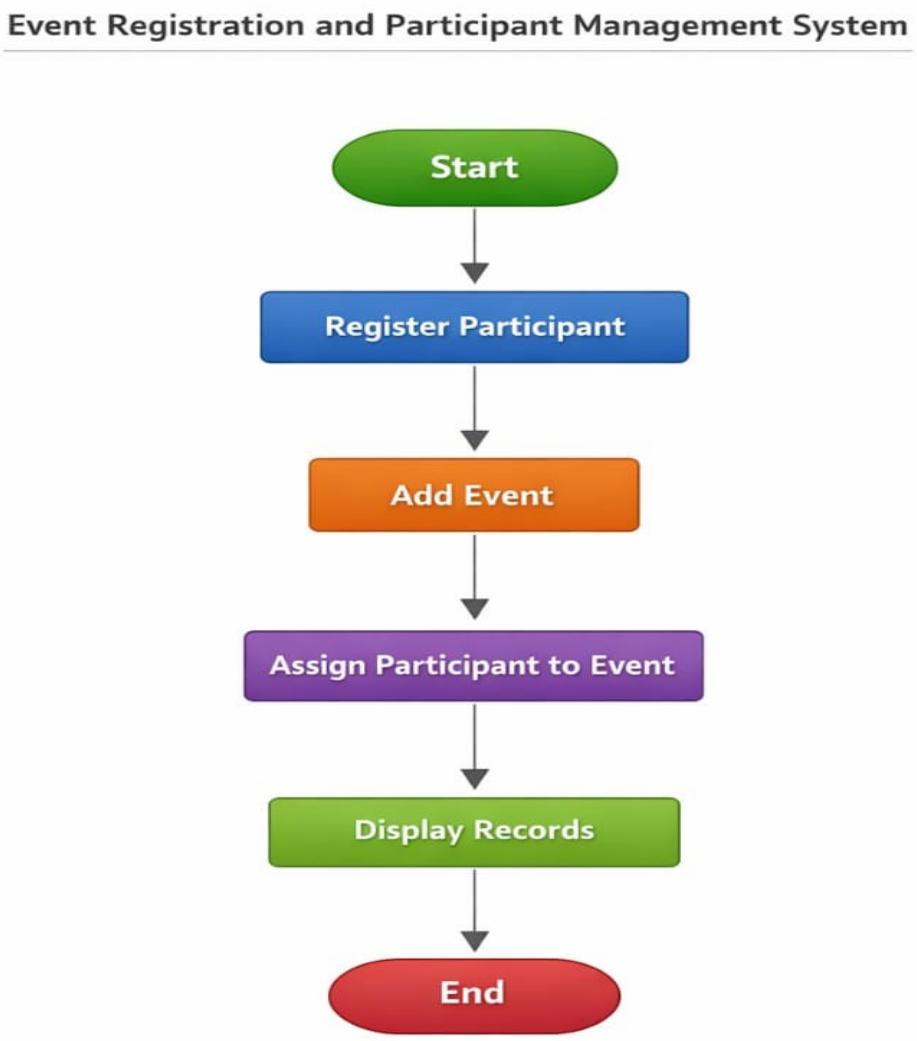
Display records

Repeat if needed

Exit program

End

4.7 Flowchart



5. SOURCE CODE

```
import java.util.*;
class Participant {
    private int id;
    private String name;
    private String email;
    public Participant(int id, String name, String email) {
        this.id = id;
        this.name = name;
        this.email = email;
    }
    public int getId() {
        return id;
    }
    public String getName() {
        return name;
    }
    public String getEmail() {
        return email;
    }
```

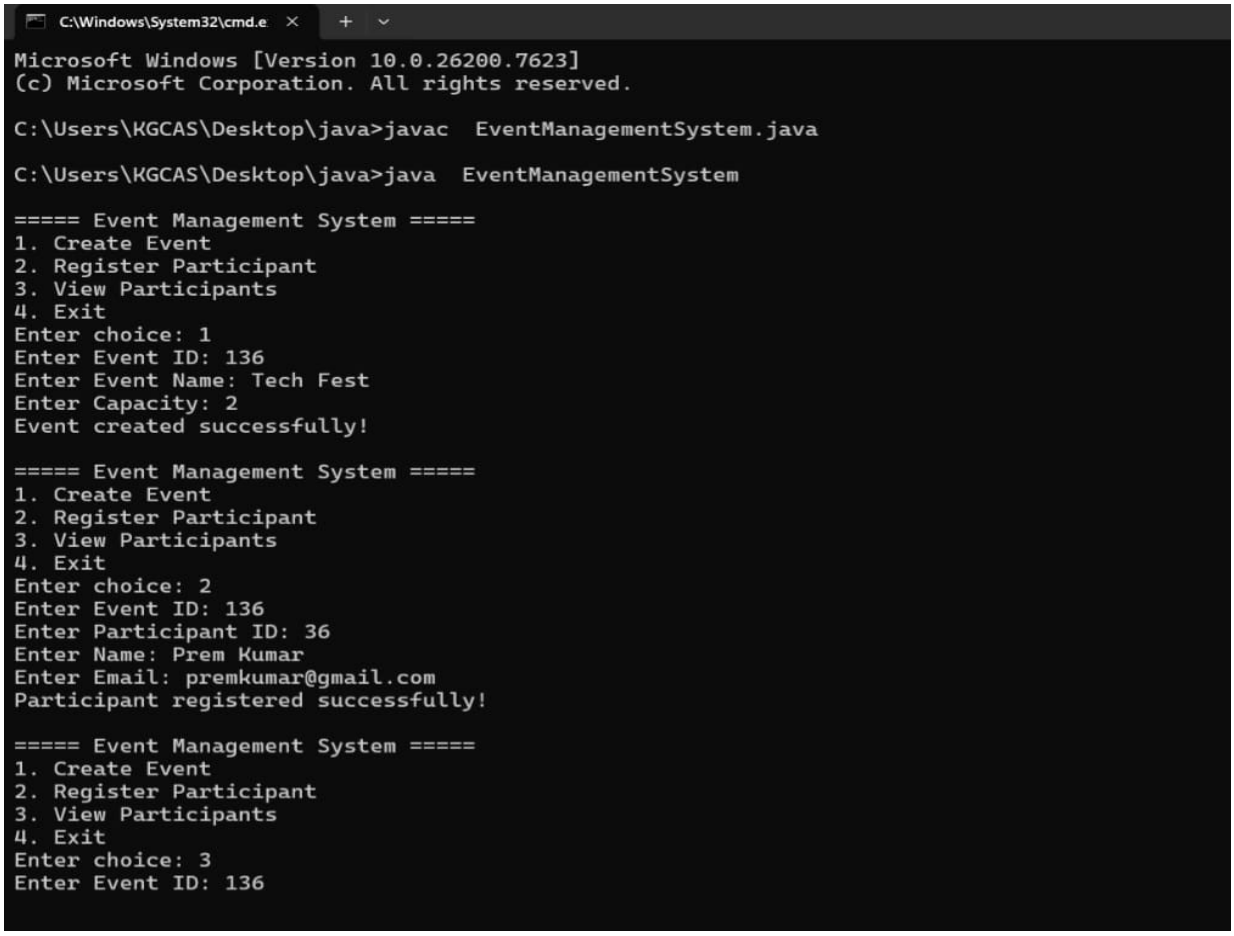
```
    public String toString() {
        return "ID: " + id + ", Name: " + name + ", Email: " + email;
    }
}
class Event {
    private int eventId;
    private String eventName;
    private int capacity;
    private List<Participant> participants;
    public Event(int eventId, String eventName, int capacity) {
        this.eventId = eventId;
        this.eventName = eventName;
        this.capacity = capacity;
        this.participants = new ArrayList<>();
    }
    public int getEventId() {
        return eventId;
    }
    public String getEventName() {
        return eventName;
    }
    public boolean addParticipant(Participant p) {
```

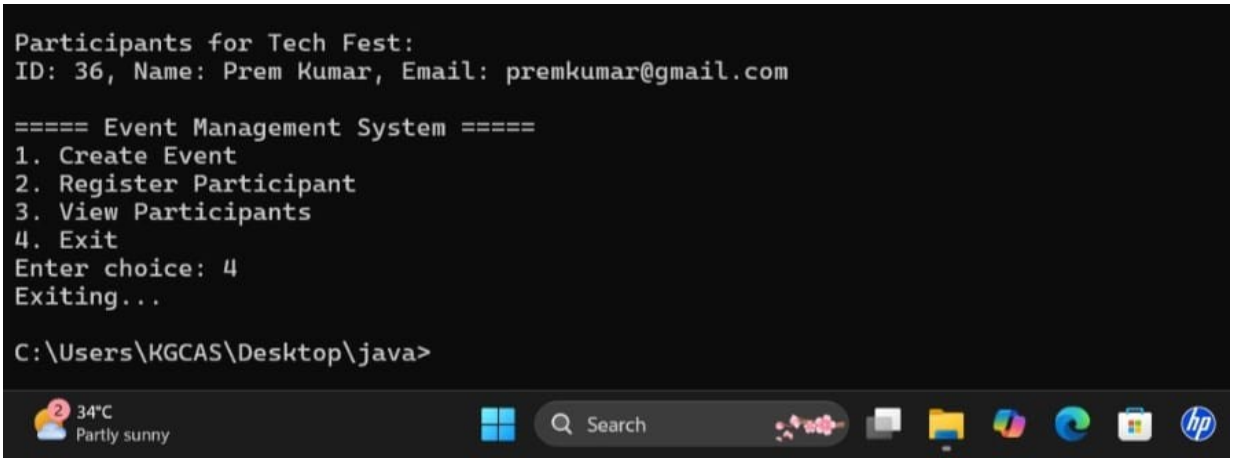
```
        if (participants.size() < capacity) {
            participants.add(p);
            return true;
        }
        return false;
    }
    public void displayParticipants() {
        if (participants.isEmpty()) {
            System.out.println("No participants registered.");
        } else {
            for (Participant p : participants) {
                System.out.println(p);
            }
        }
    }
}
public class EventManagementSystem {
    private static Map<Integer, Event> events = new HashMap<>();
    public static void main(String[] args) {
        Event event1 = new Event(1, "Tech Fest", 2);
        events.put(1, event1);
        Participant p1 = new Participant(101, "Arun", "arun@gmail.com");
```

```
        Participant p2 = new Participant(102, "Bala", "bala@gmail.com");
        Participant p3 = new Participant(103, "Cathy", "cathy@gmail.com");
        event1.addParticipant(p1);
        event1.addParticipant(p2);
        event1.addParticipant(p3); // Will not be added (capacity full)

        System.out.println("Participants for " + event1.getEventName() + ":");
        event1.displayParticipants();
    }
}
```

6. INPUT AND OUTPUT SCREENS





7. DISCUSSION & CONCLUSION

7.1 Discussion

The system was developed to simplify event registration and participant management.

It uses Core Java concepts and ensures accurate data handling.

Testing confirms that the system reduces manual effort and improves efficiency.

7.2 Conclusion

The Event Registration and Participant Management System demonstrates the use of Java programming in real-world applications.

It helps manage participant data efficiently and reduces errors.

Future enhancements may include:

• Database integration

• Online registration system

• GUI improvements

• Email notifications

8. REFERENCES

Books and Websites

• Herbert Schildt – Java: The Complete Reference

• https://www.oracle.com/java

• https://www.geeksforgeeks.org

• https://www.w3schools.com

• https://www.javatpoint.com